



**DETECTION OF MALFUNCTIONING
PHOTOVOLTAIC MODULES USING
ML AND IOT**



A PROJECT REPORT

Submitted by

DHIVYA E (613022106016)

LAVANYA R (613022106029)

NEHASRI M (613022106037)

RITHIKA M (613022106042)

In partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

VIVEKANANDHA COLLEGE OF TECHNOLOGY FOR WOMEN

NAMAKKAL- 637 205

ANNA UNIVERSITY: CHENNAI 600 025

MAY 2026



**DETECTION OF MALFUNCTIONING
PHOTOVOLTAIC MODULES USING
ML AND IOT**



A PROJECT REPORT

Submitted by

DHIVYA E (613022106016)

LAVANYA R (613022106029)

NEHASRI M (613022106037)

RITHIKA M (613022106042)

In partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

VIVEKANANDHA COLLEGE OF TECHNOLOGY FOR WOMEN

NAMAKKAL- 637 205

ANNA UNIVERSITY: CHENNAI 600 025

MAY 2026

BONAFIDE CERTIFICATE

Certified that this Project report “**DETECTION OF MALFUNCTIONING PHOTOVOLTAIC MODULE USING ML AND IOT**” is the bonafide work of **DHIVYA E (613022106016), LAVANYA R (613022106029), NEHASRI M (613022106037), RITHIKA M (613022106042)** who carried out the project work under my supervision.

SIGNATURE

Dr. S. SATHISH M.E., Ph.D.,
HEAD OF THE DEPARTMENT

Professor and Head

Department of Electronics and

Communication Engineering

Vivekanandha college of

Technology for women

Tiruchencode-637205

SIGNATURE

Mr. K. ELAMATHIM.E.,Ph.D.,
NAME OF THE SUPERVISOR

Assistant Professor

Department of Electronics and

Communication Engineering

Vivekanandha college of

Technology for women

Tiruchencode-637205

Submitted for the University Project Viva Voce held on _____

Internal Examiner

External Examiner

ACKNOWLEDGMENT

We wish to convey our thanks to our Honorable Chairman and Secretary **‘Vidhya Rathna’ Dr. M. KARUNANITHI, Pharm.,M.S.,Ph.D., D.Litt.,** and the trust members who have provided all the facilities to develop our project successfully.

We are extremely grateful to our beloved and Honorable Executive Director **Dr. S. KUPPUSWAMI, B.E., M.Sc. (Eng)., D.Eng. (France)** who gave opportunity to frame the project to the full satisfaction.

We are extremely grateful to our beloved and Honorable Chief Executive **Mr. M. CHOCKALINGAM, M.Sc., B.Ed.,** for allowing us to have maximum use of facilities to do this project.

We are extremely grateful to our Beloved and Honorable Principal **Dr. M. DEVI, B.E., M.E., Ph.D.,** for provided us an opportunity to carried out this project.

We are very thankful to our Head of Department in Electronics and Communication Engineering **Dr. S. SATHISH, M.E., Ph.D.,** and project Coordinator **Mr. K. ELAMATHIM.E.,Ph.D.,** for their valuable help and suggestion imparted us.

We are very thankful to our Project guide **Mrs.P.SHAMUGAPRIYA, M.E.,** who had motivated us and provided Guidance throughout the execution of our project.

Finally, we are very thankful to all the Teaching, Non-Teaching Staff Members, Parents, Siblings and Friends for their support, encouragement and effective ideas during this project.

ABSTRACT

The increasing demand for renewable energy has led to the widespread adoption of solar photovoltaic (PV) systems. However, the efficiency and reliability of these systems are often affected by various faults such as partial shading, dust accumulation, cell degradation, temperature variations, and electrical failures. Detecting these faults at an early stage is essential to maintain optimal performance and reduce energy losses. Traditional fault detection methods rely on manual inspection and periodic maintenance, which are time-consuming, less accurate, and not suitable for large-scale solar installations.

To overcome these limitations, this project proposes an intelligent system for the detection of malfunctioning photovoltaic modules using Machine Learning (ML) and Internet of Things (IoT) technologies. The system continuously monitors key parameters such as voltage, current, and temperature using sensors connected to a microcontroller. These real-time values are processed and transmitted through a wireless communication module (ESP8266) to an IoT platform for remote monitoring and analysis.

Machine Learning techniques, particularly Artificial Neural Networks (ANN), are employed to analyze the collected data and identify abnormal patterns that indicate faults in the PV modules. In addition, an LCD display is used to show real-time values locally, and a buzzer alert is triggered when abnormal conditions are detected.

This approach improves fault detection accuracy, reduces maintenance efforts, and enhances the overall efficiency and lifespan of photovoltaic systems. The proposed system is cost-effective, scalable, and suitable for both small-scale and large-scale solar power applications.

LIST OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	
	LIST OF FIGURES	
	LIST OF ABBREVIATION	1
1	INTRODUCTION	2
	1.1. Global Growth of Solar Energy Systems	
	1.2. Challenges in Photovoltaic System Monitoring	
	1.3. Emergence of IoT in PV Monitoring Systems	
	1.4. Role of Machine Learning in Fault Detection	
2	LITERATURE SURVEY	6
3	EXISTING SYSTEM	10
	3.1 Overview of current PV Monitoring Systems	
	3.2 Core components and Architecture	
	3.3 Operation workflow	
	3.4 Key implementation in Existing Systems	
	3.5 Strength of Existing Systems	
	3.6 Challenges and limitation	

4	SYSTEM IMPLEMENTATION	14
	4.1 Proposed system	
	4.2 Proposed block diagram	
	4.3 Block diagram description	
	4.4 Advantages of the system	
5	MODULE DESCRIPTION	20
	5.1 Hardware Requirements	
	5.1.1 Solar panel	
	5.1.2 Voltage sensor	
	5.1.3 Temperature sensor	
	5.1.4 current sensor	
	5.1.5 Arduino uno	
	5.1.6 ESP 8266	
	5.1.7 BUZZER	
	5.1.8 LCD	
	5.1.9 Battery	

5.2 Software Requirements

5.2.1 Blynk

5.2.2 Arduino IDE

5.2.3 Machine Learning Frameworks

5.2.4 IOT Communication Protocols

6	RESULT & DISCUSSION	40
7	CONCLUSION & FUTURE SCOPE	43
8	APPENDIX 1	47
9	APPENDIX 2	52
10	REFERENCES	53

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
4.1	PROPOSED BLOCK DIAGRAM	15
5.1	SOLAR PANEL	21
5.2	VOLTAGE SENSOR	23
5.3	a) NEGATIVE TEMPERATURE COEFFICIENT	25
5.3	b) POSITIVE TEMPERATURE COEFFICIENT	26
5.4	CURRENT SENSOR	28
5.5	ARUDINO UNO	29
5.6	ESP 8266	30
5.7	BUZZER	31
5.8	LCD	32
5.9	BATTERY	34
5.10	ARDUINO SOFTWARE IDE	36
5.11	COMPONENTS CONNECTION	40
5.12	OUTPUT	41

LIST OF ABBREVIATION

ADC – Analog-to-Digital Converter

ANN – Artificial Neural Network

ALM – Alarm

DC – Direct Current

GND – Ground

I/O – Input/Output

LCD – Liquid Crystal Display

LED – Light Emitting Diode

MCU – Microcontroller Unit

PCB – Printed Circuit Board

PC – Personal Computer

PV – Photovoltaic Module

RX – Receiver, TX – Transmitter

UNO – Universal Number One (Arduino UNO Board)

VCC– Voltage at Common Collector (Power Supply Voltage)

IoT – Internet of Things

CHAPTER 1

INTRODUCTION

In recent years, the growing concern over environmental pollution and the depletion of fossil fuels has led to a significant increase in the use of renewable energy sources. Among these, solar energy has emerged as one of the most promising and sustainable energy solutions. Photovoltaic (PV) systems convert sunlight directly into electrical energy and are widely used in residential, commercial, and industrial applications. However, maintaining the efficiency and reliability of PV systems remains a major challenge due to various environmental and operational factors.

Photovoltaic modules are exposed to harsh environmental conditions such as dust, shading, temperature fluctuations, and aging, which can lead to performance degradation and faults. These faults may include partial shading, hot spots, open-circuit faults, short circuits, and degradation of cells over time. If not detected early, these issues can significantly reduce energy output, increase maintenance costs, and even lead to complete system failure.

Traditional methods for monitoring PV systems involve manual inspection and basic electrical measurements. These methods are not only time-consuming but also inefficient for large-scale solar farms. Moreover, they fail to provide real-time insights and cannot detect complex or early-stage faults accurately. Therefore, there is a need for an automated, intelligent, and real-time monitoring system that can efficiently detect and diagnose faults in photovoltaic modules.

1.1 Global Growth of Solar Energy Systems

In recent years, solar energy has experienced significant growth due to increasing energy demands and environmental concerns. Governments and organizations worldwide are investing heavily in solar power projects to promote renewable energy adoption. Large-scale solar farms and rooftop installations are becoming common in both urban and rural areas.

With this rapid expansion, maintaining the performance of PV systems has become a major challenge. Even a small fault in a solar panel can lead to considerable energy loss over time. As the number of installed systems increases, the need for advanced monitoring and fault detection techniques becomes more important.

Additionally, climate change and unpredictable weather conditions further affect the performance of solar panels. Variations in sunlight intensity, temperature, and environmental conditions directly impact the output of PV systems. Hence, continuous monitoring is essential to ensure optimal performance and efficiency.

1.2 Challenges in Photovoltaic System Monitoring

Photovoltaic systems face several challenges that affect their performance and lifespan. Common issues include partial shading caused by trees or buildings, accumulation of dust and dirt on panel surfaces, cell degradation due to aging, and electrical faults such as open circuits and short circuits.

Detecting these faults manually is not practical, especially in large solar installations. Traditional methods do not provide real-time information and often fail to identify early-stage faults.

As a result, minor issues may go unnoticed and develop into major problems, leading to system failure. Another major challenge is the lack of intelligent decision-making in conventional systems. They cannot analyze multiple parameters simultaneously or predict potential failures. This highlights the need for a smart monitoring system that can automatically detect and classify faults based on real-time data.

1.3 Emergence of IoT in PV Monitoring Systems

The Internet of Things (IoT) has revolutionized the way systems are monitored and controlled. IoT enables the connection of sensors, devices, and systems through the internet, allowing real-time data collection and communication.

In photovoltaic systems, IoT is used to monitor parameters such as voltage, current, and temperature using sensors. These sensors are connected to a microcontroller, which processes the data and sends it to a cloud platform through wireless communication modules like ESP8266.

This allows users to monitor system performance remotely through dashboards and mobile applications. IoT also enables data storage and analysis, helping in understanding system behavior over time. The ability to access real-time data from anywhere improves decision-making and reduces maintenance efforts.

1.4 Role of Machine Learning in Fault Detection

Machine Learning (ML) plays a vital role in improving the accuracy and efficiency of fault detection in PV systems. ML algorithms can analyze large amounts of data and identify patterns that indicate normal and faulty conditions.

In this project, an Artificial Neural Network (ANN) is used to detect malfunctions in photovoltaic modules. The model is trained using different sets of data representing various operating conditions. Once trained, it can classify real-time data and identify abnormalities.

ML-based systems can detect faults at an early stage, even before they become severe. This helps in preventing energy loss, reducing maintenance costs, and increasing system reliability. The combination of ML with IoT creates a powerful solution for intelligent monitoring and fault detection.

1.5 Role of IoT and ML in Improving System Reliability

The integration of IoT and Machine Learning provides a smart and efficient solution for monitoring photovoltaic systems. IoT ensures continuous data collection and real-time communication, while ML enables intelligent analysis and fault detection.

This combination allows the system to automatically detect abnormal conditions, generate alerts, and provide insights into system performance. It reduces human intervention and ensures timely maintenance, thereby improving the overall efficiency and lifespan of the PV system.

The proposed system is cost-effective, scalable, and suitable for both small-scale and large-scale applications. It represents a significant step towards the development of smart and sustainable energy systems.

CHAPTER-2

LITERATURE SURVEY

1.Title:

IoT-Based Solar Panel Monitoring System

Authors: R. Verma, S. Kumar

Description:

This paper presents a basic IoT-based system for monitoring photovoltaic parameters such as voltage and current in real time. The system uses sensors and transmits data to a cloud platform for visualization. It helps users track solar panel performance remotely. However, the system does not include any Machine Learning techniques for fault detection or intelligent analysis. It only provides monitoring without identifying or classifying faults, making it less effective compared to advanced smart systems.

2.Title:

Machine Learning-Based Fault Diagnosis for Photovoltaic Systems

Authors: A. Mellit, S. Kalogirou

Description:

This research focuses on using Machine Learning algorithms to detect faults in photovoltaic systems. The model is trained using historical data to identify abnormal patterns and classify faults. It improves detection accuracy compared to traditional methods. However, the system does not include IoT integration for real-time monitoring and data transmission. It mainly works as an offline analytical model,

limiting its practical application in real-time environments.

3.Title:

Solar PV Monitoring Using Arduino and Sensors

Authors: K. Ramesh, P. Lakshmi

Description:

This paper proposes a simple hardware-based monitoring system using Arduino and sensors to measure voltage, current, and temperature. The collected data is displayed locally using an LCD. The system is easy to implement and cost-effective. However, it lacks wireless communication, IoT integration, and intelligent fault detection mechanisms. It cannot monitor data remotely or detect abnormal conditions automatically.

4. Title:

Fault Detection in PV Systems Using Threshold-Based Method

Authors: D. Patel, R. Shah

Description:

This study introduces a threshold-based fault detection method for photovoltaic systems. The system compares measured values with predefined threshold limits to identify faults. It is simple and easy to implement but lacks accuracy and flexibility. It cannot detect complex faults or adapt to changing environmental conditions.

5. Title:

ANN-Based Fault Detection in Solar Panels

Authors: S. Karthik, M. Praveen

Description:

This paper uses Artificial Neural Networks (ANN) to detect faults in solar panels based on electrical parameters. The model improves detection accuracy by learning patterns from data. However, the system is mainly simulation-based and does not include real-time hardware implementation or IoT connectivity. It lacks remote monitoring and real-time alert features, reducing its practical usability.

6. Title:

IoT-Based Energy Monitoring System for Solar Panels

Authors: J. Saranya, V. Divya

Description:

This research presents an IoT-based energy monitoring system that tracks solar panel output and sends data to an online platform. It allows users to monitor performance remotely and store historical data. However, the system focuses only on monitoring and does not include intelligent fault detection or Machine Learning analysis. It cannot identify or classify faults automatically.

7. Title:

Thermal Image-Based Fault Detection in PV Modules

Authors: H. Sharma, R. Singh

Description:

This study uses thermal imaging techniques to detect faults such as hotspots in photovoltaic modules. Infrared cameras capture temperature variations, which are analyzed to identify defective areas. While the method provides accurate results, it requires expensive equipment and complex processing. It also lacks IoT-based real-time monitoring and is not suitable for low-cost implementations.

8. Title:

Remote Monitoring of Solar Panels Using GSM Technology

Authors: P. Anitha, R. Kumar

Description:

This paper proposes a remote monitoring system using GSM technology to send solar panel data via SMS. It allows basic remote monitoring without internet connectivity. However, the system has limited data transmission capability and does not support real-time visualization or cloud storage. It also lacks Machine Learning-based fault detection and advanced analysis

CHAPTER 3

EXISTING SYSTEM

With the rapid growth of solar energy systems, monitoring and maintenance of photovoltaic (PV) modules have become critical for ensuring optimal performance. Faults such as partial shading, dust accumulation, temperature rise, and electrical failures significantly affect the efficiency of solar panels. In large-scale or remote installations, identifying malfunctioning PV modules at the right time is a challenging task.

Existing systems mainly rely on image processing techniques and machine learning models to detect faults in PV modules. These systems use thermal imaging and advanced algorithms to analyze temperature variations and identify abnormal conditions.

While they provide high accuracy, they are often complex, expensive, and not suitable for real-time low-cost applications. This chapter explains the structure, components, and working of current PV fault detection systems and highlights their advantages and limitations.

3.1 Overview of Current PV Monitoring Systems

Earlier PV monitoring systems were based on manual inspection and periodic measurements of electrical parameters like voltage and current. These methods were not efficient for large solar farms and often failed to detect early-stage faults.

Modern systems have shifted towards automated monitoring using thermal imaging and machine learning techniques. These systems capture infrared images of solar panels and analyze temperature differences to detect faults such as hotspots and damaged cells

.Although these systems improve detection accuracy, they are mostly used in industrial environments and research applications due to their complexity and cost.

3.2 Core Components and Architecture

The architecture of existing PV fault detection systems generally includes:

Sensing / Imaging Units:

- Infrared (IR) Cameras are used to capture thermal images of solar panels.
- Detect temperature variations that indicate faults such as hotspots or damaged cells.

Data Processing Units:

- High-performance computing systems process captured images.
- Preprocessing techniques include:
 - a. Gamma correction (image enhancement)
 - b. Threshold filtering (noise reduction)

Machine Learning Models:

- Convolutional Neural Networks (CNN) for image-based fault detection.
- eXtreme Gradient Boosting (XGBoost) for analyzing extracted temperature features.
- Hybrid models combining multiple algorithms for better accuracy.

Data Storage Systems:

- Local servers or cloud platforms store image data and analysis results.
- Used for training models and maintaining historical records.

User Interface:

- Software dashboards display thermal images and fault detection results.
- Used by engineers for monitoring and decision-making.

3.3 Operation Workflow

The operation of existing systems follows these steps:

Image Acquisition:

- Thermal images of PV modules are captured using IR cameras.

Preprocessing:

- Image enhancement techniques like gamma correction are applied.
- Noise is reduced using threshold functions.

Feature Extraction:

- Temperature values and statistical features are extracted from images.

Model Analysis:

- Machine learning models (CNN or XGBoost) analyze the data.
- Faults are classified based on learned patterns.

Result Interpretation:

- Results are displayed through software dashboards.
- Engineers analyze the output for maintenance actions.

3.4 Key Implementations in Existing Systems

Some commonly used approaches include:

- **CNN-Based Thermal Image Systems:**

Use deep learning models to detect faults from infrared images.

- **Hybrid Models with Preprocessing Techniques:**

Combine image enhancement methods with machine learning for improved accuracy.

- **XGBoost-Based Systems:**

Use statistical temperature data instead of images for faster processing.

These systems provide accurate results but require advanced infrastructure.

3.5 Strengths of Existing Systems

- High accuracy in detecting faults
- Ability to identify complex fault patterns
- Automated fault detection reduces manual effort
- Suitable for large-scale solar power plants
- Effective in detecting hidden faults like hotspots

3.6 Challenges and Limitations

Despite their advantages, existing systems have several limitations:

- **High Cost:** Infrared cameras and processing systems are expensive
- **Complex Implementation:** Requires skilled professionals
- **No Real-Time IoT Integration:** Limited remote monitoring capability
- **High Computational Requirements:** Needs powerful systems for processing
- **Maintenance Difficulty:** Equipment handling is complex
- **Not Cost-Effective:** Not suitable for small-scale or rural applications

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 PROPOSED SYSTEM

The proposed system presents a smart and cost-effective solution for detecting malfunctioning photovoltaic (PV) modules using the integration of Internet of Things (IoT) and Machine Learning techniques. Unlike existing methods that depend on expensive thermal imaging and complex processing, this system uses sensor-based data to monitor the performance of solar panels in real time. The main objective is to improve efficiency, reduce maintenance efforts, and enable early fault detection.

In this system, sensors are used to measure important parameters such as voltage, current, and temperature of the solar panel. These values provide key information about the operating condition of the PV module. The collected data is sent to an Arduino microcontroller, which processes the sensor inputs by converting analog signals into digital form and preparing them for further analysis.

To enable real-time remote monitoring, the system uses an ESP8266 Wi-Fi module to transmit the processed data to an IoT platform. The IoT platform displays the data through a cloud-based dashboard, allowing users to monitor system performance from anywhere. It also stores historical data, which helps in analyzing system behavior over time.

An important feature of the proposed system is the use of an Artificial Neural Network (ANN) for fault detection. The ANN model is trained with both normal and faulty data conditions. When real-time sensor data is received, the model analyzes it and identifies whether the system is operating normally or experiencing

a fault. This allows early detection of issues such as abnormal temperature rise, reduced current, or voltage fluctuations

Additionally, the system includes a local monitoring and alert mechanism. An LCD display shows real-time values of voltage, current, and temperature, while a buzzer provides an immediate alert when a fault is detected. This ensures quick response and prevents further damage to the system.

Overall, the proposed system provides a simple, efficient, and low-cost solution for photovoltaic fault detection. By combining IoT-based monitoring with machine learning, it ensures real-time performance tracking, accurate fault identification, and improved reliability, making it suitable for both small and large-scale solar applications.

4.2 PROPOSED BLOCK DIAGRAM

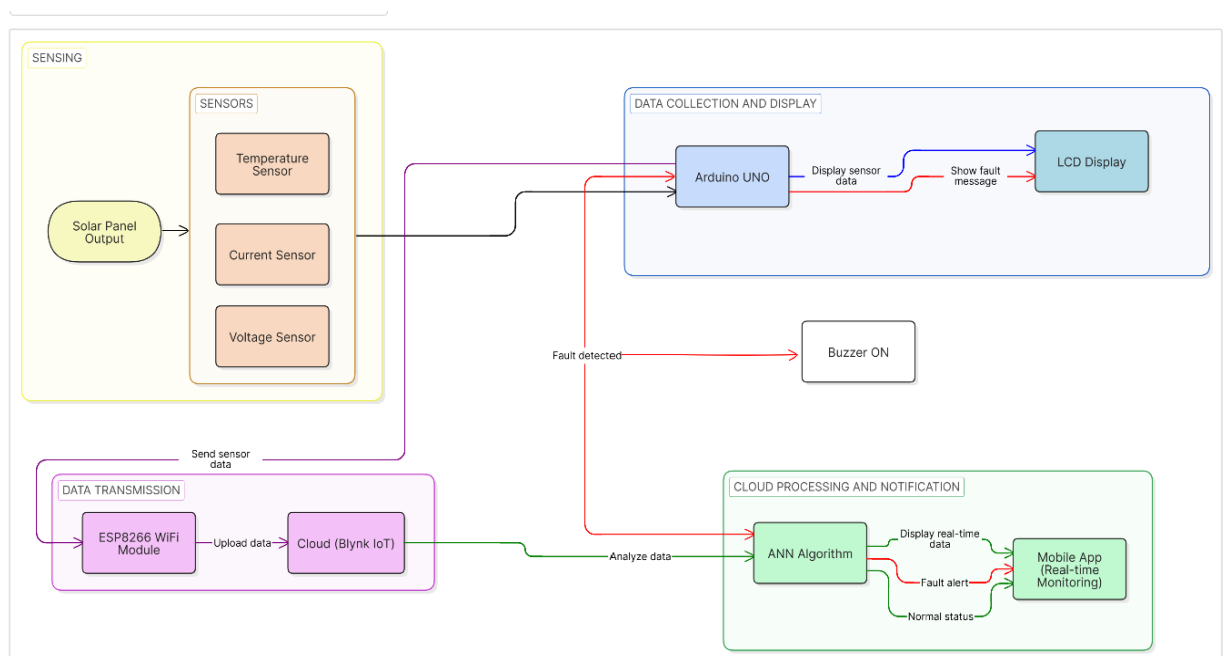


FIG 4.1: Block diagram of detecting malfunctioning PV module

4.2 BLOCK DIAGRAM DESCRIPTION

1.Solar Panel Output:

- **Power Supply:** The solar panel itself acts as a power source by converting sunlight into DC electrical energy. The 12V output is stepped down and regulated to 5V for the microcontroller (Arduino), 3.3V for Wi-Fi module (ESP8266).
- **Connection:** The output of the solar panel is connected to the voltage sensor and current sensor to measure electrical parameters, and it is also connected to the power supply circuit to regulate the voltage for the overall system.
- **Function:** It generates electrical energy and provides input power to the entire monitoring system.

2.Sensors:

- **Power Supply:** The sensors are powered using a regulated 5V supply from the Arduino UNO.
- **Connection:** The temperature sensor is connected to the solar panel to measure its temperature, the current sensor is connected in series with the panel output to measure current, and the voltage sensor is connected across the panel to measure voltage. All sensor outputs are connected to the Arduino UNO input pins.
- **Function:** The sensors measure temperature, current, and voltage of the solar panel and continuously send real-time data to the system.

3.Data Collection (Microcontroller):

- **Power Supply:** The Arduino UNO is powered by a regulated 5V DC supply.
- **Connection:** Sensor outputs are connected to the input pins of the Arduino, and it is also connected to the LCD, buzzer, and Wi-Fi module.
- **Function:** It reads and processes sensor data, checks for faults, and controls output actions like display, alert, and data transmission.

4. Display Unit:

- Power Supply: The LCD display is powered by a 5V supply from the Arduino.
- Connection: The LCD is connected to the output pins of the Arduino.
- Function: It displays real-time values and fault messages for easy monitoring.

5. Alert System:

- Power Supply: The buzzer is powered by a regulated 5V DC supply from the Arduino UNO, ensuring reliable operation during fault conditions.
- Connection: The buzzer is connected to one of the digital output pins of the Arduino UNO. When a fault is detected, the Arduino sends a signal to activate the buzzer.
- Function: The alert system plays a crucial role in providing immediate local notification to the user. This helps in quick identification of faults without the need to check the display or mobile application.

. 6. Data Transmission:

- Power Supply: The Arduino UNO is powered by a regulated 5V DC supply, which ensures stable operation of the microcontroller.
- Connection: The outputs from the temperature, current, and voltage sensors are connected to the analog and digital input pins of the Arduino UNO. These connections allow the microcontroller to receive real-time data from the solar panel. The Arduino is also connected to other components like the LCD display, buzzer, and Wi-Fi module for further processing and communication.
- Function: The Arduino UNO acts as the central processing unit of the system. It continuously reads the sensor data, converts analog signals into digital values, and processes them using programmed logic.

7. Cloud Platform (IoT):

- **Power Supply:** The cloud platform operates on remote servers and does not require any direct hardware power supply from the system.
- **Connection:** The ESP8266 module connects to the internet and uploads data to the Blynk IoT cloud platform.
- **Function:** The cloud platform stores real-time data received from the system and makes it accessible from anywhere. It acts as an interface between the hardware system and the user.

8. Data Processing (ANN Algorithm):

- **Power Supply:** The ANN algorithm runs on cloud or server-based systems, so it does not require direct hardware power supply from the circuit.
- **Connection:** The algorithm receives input data from the cloud platform where sensor data is stored. It processes this data for intelligent analysis.
- **Function:** The Artificial Neural Network (ANN) is used to analyze the sensor data and identify patterns. It detects faults such as panel degradation, abnormal temperature, or electrical issues Power.

9. Mobile Application:

- **Power Supply:** The mobile application runs on a smartphone powered by its internal battery.
- **Connection:** The mobile app connects to the cloud platform via the internet. It retrieves data uploaded by the ESP8266 module.
- **Function:** The mobile application provides a user-friendly interface for monitoring the system. It displays real-time values of temperature, voltage, and current, and sends notifications when faults occur. It allows users to check system status anytime and from anywhere, ensuring efficient remote monitoring and control.

4.4 ADVANTAGES OF THE PROPOSED SYSTEM

- 1. Low Cost Solution:** The proposed system uses simple and low-cost sensors such as voltage, current, and temperature sensors instead of expensive equipment like thermal imaging cameras. This significantly reduces the overall system cost, making it affordable and suitable for small-scale as well as large-scale applications.
- 2. Real-Time Monitoring:** The system continuously monitors the performance of the solar panel and updates the data instantly through the IoT platform. This allows users to observe system parameters like voltage, current, and temperature in real time without any delay.
- 3. Early Fault Detection:** With the help of the Artificial Neural Network (ANN) algorithm, the system can detect faults at an early stage. Early identification of issues helps in preventing serious damage to the solar panel and reduces power loss.
- 4. Reduced Maintenance Effort:** Since the system automatically monitors and detects faults, it reduces the need for manual inspection. This saves time, reduces human effort, and lowers maintenance costs.
- 5. Remote Accessibility:** Using the ESP8266 Wi-Fi Module, the system sends data to the cloud, allowing users to monitor the solar panel from anywhere through a mobile or web application. This is especially useful for remote or large installations.
- 6. Accurate Fault Identification:** The use of ANN improves the accuracy of fault detection. It can effectively distinguish between normal and faulty conditions by analyzing patterns in the data, reducing false alarms.
- 7. Data Storage and Analysis:** The cloud platform stores historical data, which can be used to analyze system performance over time. This helps in understanding trends and predicting future faults, improving system.

CHAPTER 5

MODULE DESCRIPTION

5.1 HARDWARE REQUIREMENTS

5.1.1 SOLAR PANEL (PHOTOVOLTAIC MODULE)

The solar panel is the primary energy source of the system and plays a crucial role in the detection and analysis of photovoltaic (PV) module performance. It converts sunlight directly into electrical energy using the principle of the Photovoltaic Effect. A solar panel is made up of multiple solar cells, typically composed of semiconductor materials such as silicon. When sunlight falls on these cells, photons excite electrons, generating a flow of electric current. This results in the production of direct current (DC) electricity, which is used as the input for the monitoring system.

In this project, a mini solar panel is used to simulate real-time working conditions of a photovoltaic system. The output of the panel, mainly voltage and current, varies depending on environmental factors such as sunlight intensity, temperature, and shading conditions. These variations are important indicators of the panel's health and efficiency. This helps in reducing power loss, improving efficiency, and minimizing maintenance costs. Thus, the solar panel module not only serves as an energy generator but also acts as a critical component for performance monitoring and fault detection in the overall system.

Types of Solar Panels

- Monocrystalline Solar Panel
- Polycrystalline Solar Panel
- Thin-Film Solar Panel
- Amorphous Silicon Solar Panel

i) polycrystalline solar panel

A polycrystalline solar panel is a type of photovoltaic (PV) panel made from multiple silicon crystals melted together and poured into a mold to form solar cells. Unlike single-crystal panels, these cells have a fragmented crystal structure, which gives them a distinctive bluish color with a grainy appearance. The working of a polycrystalline solar panel is based on the Photovoltaic Effect, where sunlight falling on the panel generates electrical energy. When photons from sunlight strike the silicon material, electrons are excited and produce a flow of electric current. This results in the generation of direct current (DC) electricity.

Polycrystalline panels typically have a moderate efficiency ranging from 13% to 17%, which is slightly lower than monocrystalline panels. This is due to the presence of multiple crystal boundaries that restrict the free movement of electrons. However, they are widely preferred because of their lower manufacturing cost and simpler production process. These panels are highly suitable for small-scale applications, academic projects, and residential systems where cost is an important factor. In this project, the polycrystalline solar panel is used to generate electrical energy, and its output parameters such as voltage and current are continuously monitored using sensors. Any deviation in these parameters helps in identifying faults or malfunctioning conditions in the panel.

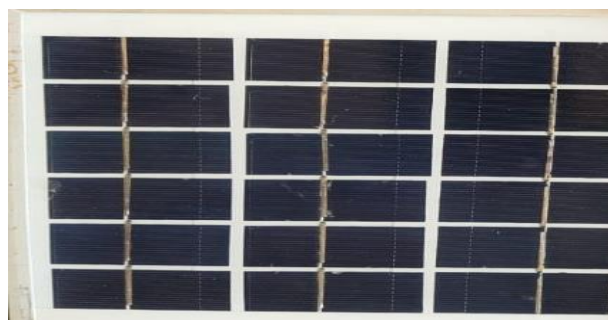


Fig 5.1 Solar panel

5.1.2 VOLTAGE SENSOR

The voltage sensor is an essential component in this system, used to continuously monitor the output voltage of the solar panel. It plays a key role in analyzing the performance and health of the photovoltaic module by providing accurate real-time voltage data. Solar panels generate DC voltage that can vary depending on environmental conditions such as sunlight intensity, temperature, and load variations. In some cases, the generated voltage may exceed the safe input range of the microcontroller.

The voltage divider circuit consists of resistors that proportionally scale down the input voltage. This reduced voltage is then fed into the analog input pin of the Arduino UNO, where it is converted from analog to digital form using the built-in Analog-to-Digital Converter (ADC). The Arduino processes this digital value and calculates the actual voltage of the solar panel using appropriate scaling formulas.

Continuous monitoring of voltage is very important for detecting abnormal conditions in the solar panel. Under normal operation, the voltage remains within a predictable range. However, any sudden drop, fluctuation, or irregular pattern in voltage may indicate issues such as partial shading, dust accumulation on the panel surface, loose electrical connections, or even internal damage to the solar cells.

The voltage data collected by the sensor is further used in the system for IoT-based monitoring and Machine Learning analysis. By comparing real-time voltage values with expected patterns, the system can identify faults at an early stage and take necessary actions, such as triggering alerts or notifying users through the cloud platform. Thus, the voltage sensor not only ensures safe interfacing between the solar panel and the microcontroller.

i) Voltage Divider Type

This sensor works on the principle of a voltage divider, where two resistors are connected in series to reduce the high input voltage into a lower, proportional voltage that is safe for the microcontroller. It is used to measure the output voltage of the solar panel. Since the solar panel can generate varying voltage levels depending on sunlight intensity, directly connecting it to the microcontroller may cause damage.

Therefore, the voltage sensor scales down the voltage and sends it to the analog input pin of the Arduino UNO. The Arduino then converts this analog signal into digital form using its built-in ADC and calculates the actual voltage using a scaling factor.

Continuous monitoring of voltage is very important in this system, as any sudden drop, fluctuation, or abnormal value may indicate faults such as shading, dust accumulation, loose connections, or panel damage.

Thus, the resistive voltage sensor ensures safe measurement and plays a crucial role in fault detection and performance monitoring of the photovoltaic system.

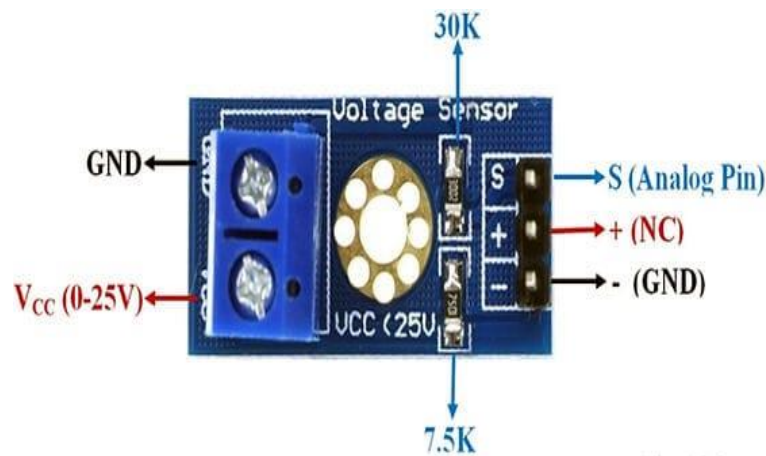


Fig 5.2 Voltage sensor

5.1.3 TEMPERATURE SENSOR

Temperature Sensor such as the LM35 Temperature Sensor or DHT11 Temperature Sensor, is used to measure the temperature of the solar panel or its surroundings. It provides an analog or digital output proportional to temperature. The LM35 gives an analog voltage directly in °C, making it easy to interpret. In this system, the sensor is connected to the Arduino, which reads and processes the temperature in real time. Temperature monitoring is important because high temperature reduces solar panel efficiency and may indicate abnormal conditions. When combined with voltage and current data, it helps in early fault detection and improves overall system performance and reliability. Thermistors are special types of resistors whose resistance changes significantly with temperature. They are widely used in temperature sensing and protection circuits. Thermistors are mainly classified into two types based on their temperature-resistance characteristics: NTC (Negative Temperature Coefficient) and PTC (Positive Temperature Coefficient).

i)NTC (Negative Temperature Coefficient)

An NTC (Negative Temperature Coefficient) thermistor, such as the NTC Thermistor, is a temperature-sensitive resistor in which the resistance decreases as temperature increases. It is typically made from semiconductor materials like metal oxides. At lower temperatures, the number of free charge carriers (electrons) inside the material is limited, resulting in higher resistance and reduced current flow. When the temperature rises, the thermal energy supplied to the material causes more electrons to break free from their atomic bonds, thereby increasing the number of charge carriers available for conduction. This leads to improved conductivity and a corresponding decrease in resistance. Due to this inverse relationship between temperature and resistance, NTC thermistors are highly sensitive and are widely used in temperature measurement, temperature compensation circuits, and monitoring applications. They provide a predictable change in resistance with

temperature, making them suitable for use in embedded systems and electronic circuits where accurate temperature sensing is required.



Fig 5.3 a) Negative Temperature Coefficient

ii) PTC (Positive Temperature Coefficient)

A PTC (Positive Temperature Coefficient) thermistor, such as the PTC Thermistor, is a temperature-sensitive resistor in which the resistance increases as temperature increases. It is commonly made from ceramic materials like barium titanate, which exhibit a unique electrical property depending on temperature. At normal or lower temperatures, the material has a relatively low resistance, allowing current to pass through easily.

However, as the temperature rises and reaches a specific limit known as the Curie temperature, the internal crystal structure of the material undergoes a change. This structural change significantly reduces the mobility of charge carriers, causing a rapid and sharp increase in resistance. Because of this behavior, the PTC thermistor naturally limits the current flowing through the circuit when the temperature becomes too high.

This self-regulating property makes it highly useful in protection applications such as overcurrent protection, resettable fuses, motor protection, and safety circuits. In addition to protection, PTC thermistors are also used in heating elements and temperature control systems where controlled resistance variation is required.



Fig 5.3 b)Positive Temperature Coefficient

5.1.4 CURRENT SENSOR

The current sensor, commonly the ACS712 Current Sensor, is used to measure the electric current flowing from the solar panel to the load. It operates based on the Hall Effect principle, where the magnetic field generated by the flow of current through a conductor is detected and converted into a proportional analog voltage output. This method allows the sensor to measure current without direct electrical contact, providing electrical isolation and enhancing safety between the high-power circuit and the microcontroller such as Arduino. The ACS712 produces an analog output voltage that varies according to both the magnitude and direction of the current, which is then read by the Arduino's analog pins and converted into current values using calibration formulas. Continuous current monitoring helps in evaluating the real-time performance of the solar panel. Any variation in current can indicate system behavior, such as reduced current due to shading, dust accumulation, or panel degradation, and a zero or sudden drop in current may indicate wiring

issues, disconnection, or panel failure. Thus, current sensing is essential for performance analysis, fault detection, and efficient operation of photovoltaic systems.

Types of Current Sensors

1. Hall Effect Current Sensors
2. Shunt Resistor Type Current Sensors
3. Current Transformer (CT) Sensors

i) Hall Effect Current Sensors

The ACS712 Current Sensor to measure the current flowing from the solar panel to the load. The ACS712 is based on the Hall Effect principle, which allows it to sense current without making direct electrical contact with the circuit. This provides electrical isolation between the high-power solar circuit and the low-power Arduino system, ensuring safe operation.

The sensor detects the magnetic field generated when current passes through its internal conductor and converts it into a proportional analog voltage output. This output varies according to the magnitude and direction of the current. The analog signal is then fed into the Arduino's analog input pin, where it is processed and converted into current values using calibration formulas.

In this system, the current sensor plays a vital role in monitoring the performance of the solar panel in real time. It helps identify abnormal conditions such as reduced current due to shading, dust accumulation, or panel degradation. A sudden drop or zero current reading may indicate wiring issues, disconnection, or complete panel failure.

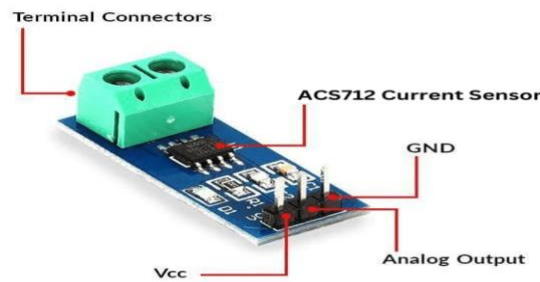


Fig 5.4 Current Sensor

5.1.5 ARDUINO UNO

The Arduino Uno is a widely used microcontroller board designed for building digital devices and interactive projects. It is based on the ATmega328P microcontroller and acts as the central processing unit of an embedded system. The board includes digital input/output pins, analog input pins, a USB interface for programming, a power jack, a reset button, and an onboard voltage regulator.

It operates at 5V and runs at a clock speed of 16 MHz, which makes it suitable for handling real-time control tasks in small and medium-scale applications. The Arduino Uno features 14 digital input/output pins, out of which 6 pins support PWM (Pulse Width Modulation) output, and 6 analog input pins used for reading sensor values such as voltage, current, temperature, and light intensity. The analog signals are converted into digital form using the built-in Analog-to-Digital Converter (ADC) with 10-bit resolution.

Additionally, the board supports communication protocols like UART, SPI, and I2C, allowing it to interface with external devices such as sensors, displays, and communication modules, making it highly versatile for embedded system designs. In terms of functionality, the Arduino Uno acts as the control unit of the system. It

continuously reads data from connected sensors, processes the information based on the uploaded program, and makes decisions accordingly.



Fig 5.5 Arduino UNO

5.1.6 ESP8266

The ESP8266 Wi-Fi Module is a low-cost, highly integrated wireless communication module used to enable Wi-Fi connectivity in embedded and IoT-based systems. It contains a built-in microcontroller with a 32-bit processor, which allows it to perform networking tasks independently or alongside another microcontroller such as the Arduino Uno. It includes GPIO pins, UART communication, and other peripherals, allowing it to interact with external hardware when required.

The ESP8266 is commonly used in two modes: Station mode (STA), where it connects to an existing Wi-Fi network, and Access Point mode (AP), where it creates its own Wi-Fi network. In many embedded systems, it is connected to a microcontroller like Arduino Uno through serial communication (UART). Since Arduino operates at 5V and ESP8266 operates at 3.3V, proper voltage level shifting or regulation is necessary to prevent damage.

The module receives data from the Arduino and transmits it over Wi-Fi to cloud platforms, servers, or mobile applications for remote monitoring and data logging. In terms of functionality, the ESP8266 plays a key role in IoT applications by enabling real-time data transmission over the internet. In a monitoring or fault detection system, it collects processed data from the Arduino and sends it to an online dashboard, allowing users to view system parameters such as voltage, current, or temperature remotely.



Fig 5.6 ESP8266

5.1.7 Buzzer

The Active Buzzer Module is an electronic output device used to produce sound as an alert or indication in embedded systems. It is commonly used in microcontroller-based projects to generate audible signals when certain conditions are met. The buzzer operates by converting electrical signals into sound waves, and in the case of an active buzzer, it contains an internal oscillator circuit, which allows it to produce sound simply by applying a DC voltage without requiring an external signal frequency. It typically operates within a voltage range of 3.3V to 5V, making it compatible with microcontrollers like the Arduino Uno.

In practical applications, the buzzer is connected to a digital output pin of a microcontroller. When the microcontroller sends a HIGH signal, the buzzer is activated and produces a continuous beep sound; when the signal is LOW, it turns off. The buzzer is a simple yet effective component for user interaction and safety alerts. It is lightweight, inexpensive, and easy to integrate into circuits without complex interfacing requirements. In combination with microcontrollers, it enhances the responsiveness of a system by providing real-time audio feedback. Due to these advantages, buzzers are widely used in alarms, timers, security systems, industrial machines, and IoT-based monitoring applications where instant attention



is required.

Fig 5.7 Buzzer

5.1.8 LCD

The 16x2 LCD Display Module is an alphanumeric display widely used in embedded systems to present real-time information in a clear and readable format. It consists of 16 columns and 2 rows, allowing it to display up to 32 characters at a time. The display is typically based on the HD44780 controller, which manages the internal operation of the LCD. It operates at 5V and consumes low power, making it suitable for microcontroller-based applications such as Arduino projects. The

LCD can display letters, numbers, and special characters, which makes it ideal for showing sensor data and system messages.

In terms of interfacing, the LCD can operate in either 4-bit or 8-bit mode. In 8-bit mode, it uses more data pins, while in 4-bit mode, it uses fewer pins by sending data in two halves. To further simplify connections, an I2C interface module is often used, which reduces the wiring to just two communication lines (SDA and SCL) along with power and ground. The microcontroller sends commands and data to the LCD, which then interprets and displays the characters accordingly.

This communication allows the system to dynamically update the displayed values based on real-time inputs from sensors. In practical applications, the 16x2 LCD is used to display important parameters such as voltage, current, temperature, and system status. In a monitoring or fault detection system, it provides immediate visual feedback to the user without requiring any external device.



Fig 5.7 LCD Display

5.1.9 BATTERY

The battery is a key power source used in embedded and IoT-based systems to provide a stable and portable supply of electrical energy to all connected components. In projects involving microcontrollers like the Arduino Uno, the battery typically supplies input power through the Vin pin or a power jack. It ensures that the system can operate independently without relying on a direct mains power

supply. Batteries are selected based on the required voltage and current of the circuit, commonly ranging from 9V batteries, lithium-ion cells, or rechargeable battery packs.

The main function of the battery is to provide consistent DC voltage to the system components such as the microcontroller, sensors, display modules like the 16x2 LCD Display Module, communication modules like the ESP8266 Wi-Fi Module, and output devices like the Active Buzzer Module. Since many electronic components require regulated voltage levels, voltage regulators or onboard regulators (like the one in Arduino Uno) are used to maintain a stable supply. This helps prevent fluctuations that could damage the components or affect system performance. In embedded systems, batteries make the setup portable and suitable for real-time and remote applications such as IoT-based monitoring systems.

They are especially useful in situations where continuous power supply is required without interruption, such as in solar-powered systems or field-based installations. Rechargeable batteries are often preferred for long-term use as they can be recharged multiple times, reducing maintenance and operational costs. Overall, the battery plays a crucial role in ensuring uninterrupted, reliable, and portable operation of the entire system.

i) Lithium-Ion (Li-ion) battery

The Lithium-Ion (Li-ion) battery is a rechargeable battery widely used in embedded systems and portable electronic devices due to its high energy density and stable output. It typically provides a nominal voltage of 3.7V per cell and can be used individually or combined in series/parallel configurations to achieve the required voltage and capacity. Li-ion batteries are commonly used in projects involving the Arduino Uno, especially when a portable and long-lasting power

source is needed. They are lightweight compared to traditional batteries and can store a large amount of energy in a small size.

One of the main advantages of Lithium-Ion batteries is their rechargeability, allowing them to be reused multiple times, which makes them cost-effective in the long run. They also have a low self-discharge rate, meaning they retain charge for a longer period when not in use. In embedded systems, Li-ion batteries are often paired with charging and protection circuits to prevent overcharging, over-discharging, and short circuits, ensuring safe operation. This is particularly important when powering modules like the ESP8266 Wi-Fi Module, 16x2 LCD Display Module, and Active Buzzer Module, which require stable voltage levels.



Fig 5.9 Lithium-Ion batteries

5.2 SOFTWARE REQUIREMENT

5.2.1 BLYNK

Blynk is a Platform with iOS and Android apps to control Arduino, Raspberry Pi, and the likes over the Internet. It's a digital dashboard where you can build a graphic interface for your project by simply dragging and dropping widgets.

Blynk is a comprehensive software suite that enables the prototyping, deployment,



and remote management of connected electronic devices at any scale. Whether it's personal IoT projects or commercial connected products in the millions, Blynk empowers users to connect their hardware to the cloud and create iOS, Android, and web applications, analyze real-time and historical data from devices, remotely control them from anywhere, receive important notifications, and much more.

Blynk was designed for the Internet of Things. It can control hardware remotely, it can display sensor data, it can store data, visualize it, and do many other cool things. It's a digital dashboard where you can build a graphic interface for your project by simply dragging and dropping widgets. It's really simple to set everything up and you'll start tinkering in less than 5 mins. Blynk is not tied to some specific board or shield. Instead, it's supporting hardware of your choice. Whether your Arduino or Raspberry Pi is linked to the Internet over Wi-Fi, Ethernet, or this new ESP8266 chip, Blynk will get you online and ready for the Internet of Your Things.

5.2.2 ARDUINO IDE

Arduino IDE (Integrated Development Environment) is a widely used open-source software platform for programming microcontrollers such as the NodeMCU ESP8266 used in this project. It provides a simple and user-friendly interface that allows developers to write, compile, and upload code to embedded devices

efficiently.

The Arduino IDE supports programming in Embedded C/C++, which is a simplified version of standard C/C++. It includes a built-in code editor with features such as syntax highlighting, auto-indentation, and error detection, making it easier for users to write and debug code. The IDE also provides a “Verify” function to compile the code and detect errors before uploading it to the microcontroller.

One of the key features of Arduino IDE is its library support. It offers a wide range of pre-built libraries that simplify interfacing with sensors, communication modules, and other hardware components. In this project, libraries are used to interface temperature sensors, voltage sensors, and Wi-Fi modules for data transmission. Another important feature is the Board Manager, which allows users to install and configure different development boards such as NodeMCU ESP8266.



```
NodeMCU
/* Comment this out to disable prints and save space */
#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

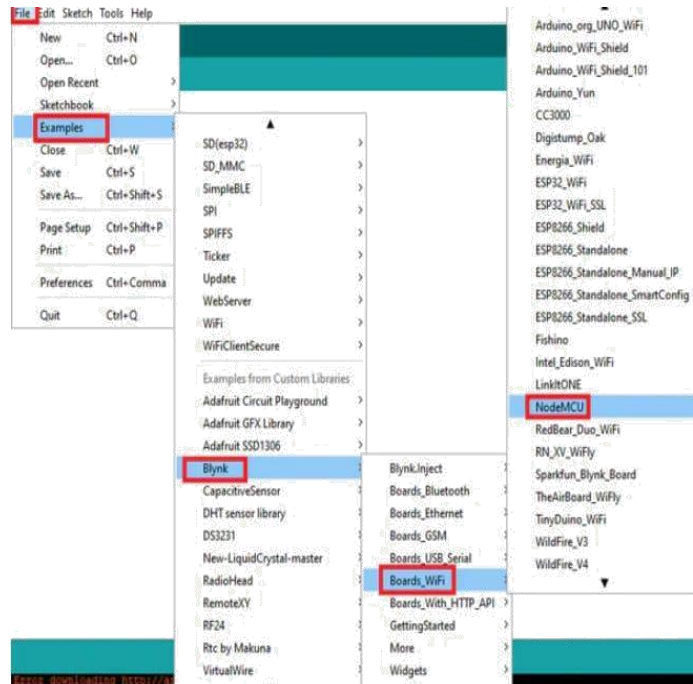
// You should get Auth Token in the Blynk App.
// Go to the Project Settings (nut icon).
char auth[] = "YourAuthToken";

// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "YourNetworkName";
char pass[] = "YourPassword";

void setup()
{
  // Debug console
  Serial.begin(9600);
```

The IDE also includes a Serial Monitor, which is used to display real-time data received from the microcontroller. This helps in debugging and verifying sensor readings during system development. The Arduino IDE plays a critical role in

firmware development by enabling smooth communication between hardware and software. It is responsible for collecting sensor data, processing it, and transmitting it to cloud platforms using Wi-Fi.



Overall, Arduino IDE is an essential tool in this project as it provides a reliable environment for coding, testing, and deploying embedded applications. Its simplicity, flexibility, and extensive community support make it ideal for IoT-based systems like photovoltaic fault detection.

5.2.3 MACHINE LEARNING FRAMEWORKS

Machine Learning (ML) frameworks are a crucial part of this project, as they enable intelligent fault detection and prediction in photovoltaic (PV) systems.

These frameworks allow the system to analyze large amounts of sensor data and identify patterns that indicate faults or performance degradation.

Python is used as the primary programming language for implementing machine learning models due to its simplicity and extensive ecosystem. Various frameworks such as TensorFlow, Keras, Scikit-learn, and XGBoost are utilized for different tasks. TensorFlow and Keras are widely used for developing deep learning models, particularly Convolutional Neural Networks (CNNs). These models are useful for analyzing complex data such as thermal images and detecting faults like hotspots, cracks, and shading in solar panels.

Scikit-learn is used for traditional machine learning algorithms such as classification and regression. It provides efficient tools for data preprocessing, feature extraction, and model evaluation. XGBoost, an advanced gradient boosting algorithm, is used for accurate fault classification based on sensor data such as voltage, temperature, and irradiance.

The machine learning process involves several steps, including data collection, preprocessing, feature selection, model training, and testing. Historical data from PV systems is used to train models, which learn to differentiate between normal and faulty conditions. Once trained, these models are deployed to analyze real-time data received from sensors. If any abnormal pattern is detected, the system identifies it as a fault and triggers alerts through the IoT platform. Overall, machine learning frameworks enhance the system's ability to perform automated fault detection, improve accuracy, and enable predictive maintenance, thereby increasing the efficiency and reliability of photovoltaic systems.

5.2.4 IoT COMMUNICATION PROTOCOLS

IoT communication protocols are essential for enabling data exchange between

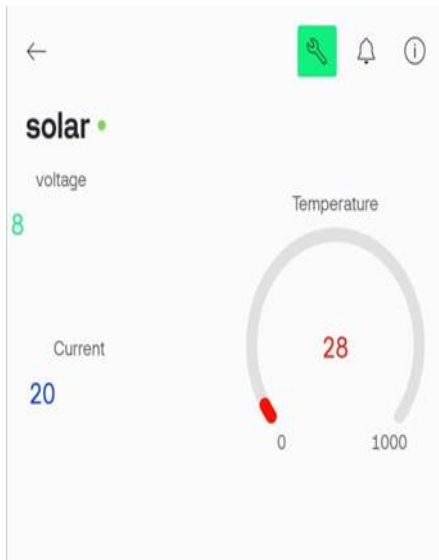
hardware devices, cloud platforms, and user interfaces in real time. In this project, protocols such as MQTT and HTTP are used to ensure reliable and efficient communication.

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol designed for IoT applications. It operates on a publish-subscribe model, where the NodeMCU publishes sensor data to a broker, and subscribed clients (such as mobile apps) receive the data. MQTT is highly efficient, consumes low bandwidth, and is ideal for real-time monitoring systems.

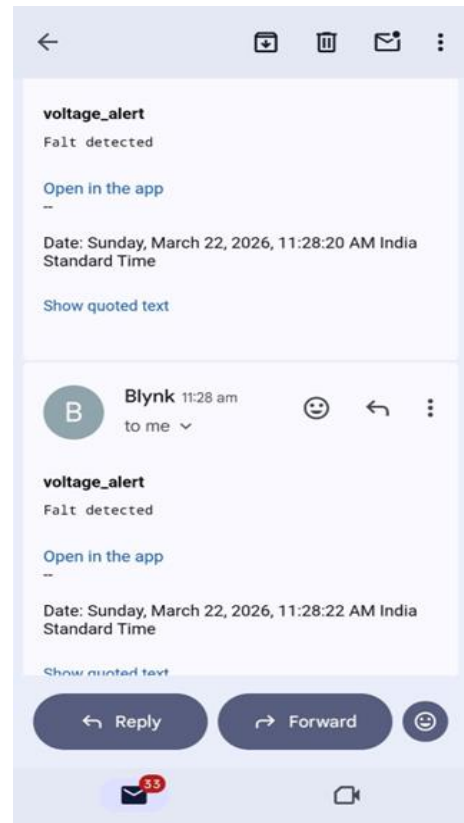
HTTP (Hypertext Transfer Protocol) is another widely used communication protocol that enables data transfer over the internet. It is used for sending sensor data to cloud servers and retrieving information for display. Although HTTP is less efficient than MQTT, it is simple and widely supported.

These protocols ensure that data collected from sensors is transmitted securely and accurately to cloud platforms such as Blynk. They support real-time updates, allowing users to monitor system performance instantly. Security is an important aspect of IoT communication. Encryption and authentication mechanisms are used to protect data from unauthorized access and ensure safe transmission.

In this project, IoT communication protocols play a key role in connecting hardware and software components, enabling seamless data flow, remote monitoring, and real-time fault detection in photovoltaic system.



a. Real time monitoring



c. Email alert notification



b. Fault detected

6.3 DISCUSSION

The results obtained from the solar monitoring system demonstrate its effectiveness in real-time data acquisition and fault detection. Under normal operating conditions, the system accurately displays voltage, current, and temperature values through the Blynk interface, allowing the user to continuously monitor system performance. The stable readings observed (8V voltage, 20A current, and 28°C temperature) indicate that the system operates efficiently within safe limits.

A key feature of the system is its ability to detect abnormal conditions. When the voltage drops significantly to 1V, the system identifies this deviation from the predefined threshold and triggers an alert notification. This confirms that the fault detection mechanism is functioning correctly. The alert message helps the user quickly recognize issues related to the solar system, reducing the risk of prolonged malfunction or damage.

Additionally, the temperature readings remain relatively constant during both normal and fault conditions, suggesting that the fault is isolated to the electrical parameter (voltage) rather than thermal factors. This highlights the system's capability to distinguish between different types of faults based on sensor data.

The integration with the Blynk platform enhances user accessibility by providing a simple and interactive interface for monitoring and notifications. However, the system could be further improved by incorporating additional parameters such as power, efficiency, or battery status, as well as implementing automated corrective actions.

Overall, the system proves to be reliable for basic solar monitoring and fault detection, making it suitable for small-scale renewable energy applications.

CHAPTER 7

CONCLUSION & FUTURE ENHANCEMENT

CONCLUSION

In this project, an intelligent system for detecting malfunctioning photovoltaic (PV) modules using Machine Learning and Internet of Things (IoT) technologies has been successfully designed and implemented. The system addresses the limitations of traditional fault detection methods by providing an automated, real-time, and efficient solution for monitoring solar panel performance.

The integration of sensors with the NodeMCU ESP8266 microcontroller enables continuous collection of important parameters such as voltage, temperature, and power output. This data is transmitted to a cloud platform, where it is analyzed and visualized using IoT applications. The use of the Blynk platform allows users to monitor system performance remotely and receive instant notifications in case of abnormal conditions.

Machine learning algorithms play a significant role in improving the accuracy and reliability of fault detection. By analyzing historical and real-time data, the system is capable of identifying different types of faults such as partial shading, overheating, and electrical failures. It reduces the need for manual inspection, enhances operational efficiency, and ensures optimal energy generation.

Although the system has certain limitations such as dependency on internet connectivity and the need for quality training data, its advantages significantly outweigh these challenges.

With further improvements, such as advanced AI models and edge computing, the system can be made even more robust and efficient.

In conclusion, the proposed solution provides a smart and effective approach for monitoring and maintaining photovoltaic systems. It contributes to improving the performance, reliability, and lifespan of solar energy systems, thereby supporting the global transition towards sustainable and renewable energy systems.

FUTURE ENHANCEMENT

Although the proposed solar monitoring system performs effectively in real-time monitoring and fault detection, several improvements can be implemented to enhance its functionality and performance.

One major enhancement would be the integration of additional parameters such as power, energy consumption, battery status, and system efficiency. This would provide a more comprehensive analysis of the solar system's performance. The inclusion of data logging and cloud storage can also be beneficial, allowing users to track historical data and analyze trends over time.

The system can be further improved by implementing automated control mechanisms. For example, in the event of a fault condition, the system could automatically disconnect the load or switch to a backup power source, reducing the risk of damage. Incorporating machine learning algorithms could also enable predictive maintenance by identifying patterns and forecasting potential failures before they occur.

Another enhancement would be the development of a more advanced user interface with graphical representations such as charts and performance analytics, making the system more user-friendly and informative. Additionally, integrating SMS, email, or multi-platform notifications can ensure that alerts reach the user even when the mobile application is not actively in use.

Finally, the system can be scaled for larger applications such as industrial or grid-level solar systems by improving sensor accuracy, communication reliability, and overall system robustness.

In conclusion, these enhancements would make the system more intelligent, reliable, and suitable for advanced solar energy management applications. Additionally, integrating SMS, email, or multi-platform notifications can ensure that alerts reach the user even when the mobile application is not actively in use.

This would provide a more comprehensive analysis of the solar system's performance. The inclusion of data logging and cloud storage can also be beneficial, allowing users to track historical data and analyze trends over time.

APPENDIX 1

SOURCE CODE

```
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
```

```
// Blynk credentials
char auth[] = "YourAuthToken";
char ssid[] = "YourWiFiName";
char pass[] = "YourPassword";
```

```
// Pin definitions
#define voltagePin A0
#define currentPin D1
#define tempPin D2
```

```
// Variables
float voltage = 0.0;
float current = 0.0;
float temperature = 0.0;
```

```
// Thresholds
float voltageThreshold = 3.0;
float tempThreshold = 50.0;
```

```
// Timer
BlynkTimer timer;
```

```
// Function to read voltage
float readVoltage()
{
    int raw = analogRead(voltagePin);
    float v = (raw / 1023.0) * 12.0;
    return v;
}
```

```
// Function to read current
float readCurrent()
{
    int state = digitalRead(currentPin);
```



```

    float c = state * 20.0;
    return c;
}

// Function to read temperature
float readTemperature()
{
    int state = digitalRead(tempPin);
    float t = state * 30.0;
    return t;
}

// Read all sensors
void readSensors()
{
    voltage = readVoltage();
    current = readCurrent();
    temperature = readTemperature();
}

// Send data to Blynk
void sendData()
{
    Blynk.virtualWrite(V0, voltage);
    Blynk.virtualWrite(V1, current);
    Blynk.virtualWrite(V2, temperature);
}

// Fault detection
void checkFault()
{
    if (voltage < voltageThreshold)
    {
        Blynk.logEvent("voltage_alert", "Fault detected: Low Voltage");
    }

    if (temperature > tempThreshold)
    {
        Blynk.logEvent("temp_alert", "Warning: High Temperature");
    }
}

// Display data on serial monitor

```

```

void printData()
{
  Serial.print("Voltage: ");
  Serial.print(voltage);
  Serial.println(" V");

  Serial.print("Current: ");
  Serial.print(current);
  Serial.println(" A");

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  Serial.println("-----");
}

// Setup
void setup()
{
  Serial.begin(9600);

  pinMode(voltagePin, INPUT);
  pinMode(currentPin, INPUT);
  pinMode(tempPin, INPUT);

  Blynk.begin(auth, ssid, pass);

  timer.setInterval(1000L, readSensors);
  timer.setInterval(1000L, sendData);
  timer.setInterval(1500L, checkFault);
  timer.setInterval(2000L, printData);
}

// Loop
void loop()
{
  Blynk.run();
  timer.run();
}

// Additional feature: Manual reset via Blynk button
BLYNK_WRITE(V3)

```

```

{
  int buttonState = param.asInt();
  if (buttonState == 1)
  {
    Serial.println("System Reset Triggered");
    voltage = 0;
    current = 0;
    temperature = 0;
  }
}

```

```

// Additional feature: LED indicator
#define ledPin D5

```

```

void controlLED()
{
  if (voltage < voltageThreshold)
  {
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    digitalWrite(ledPin, LOW);
  }
}

```

```

// Extended setup for LED
void setupLED()
{
  pinMode(ledPin, OUTPUT);
}

```

```

// Extended initialization
void extendedSetup()
{
  setupLED();
  timer.setInterval(1000L, controlLED);
}

```

```

// Call extended setup
void initializeSystem()
{
  extendedSetup();
}

```

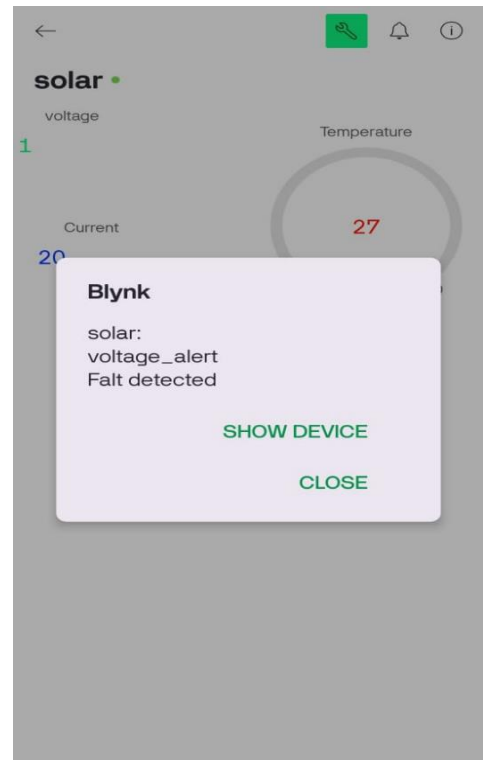
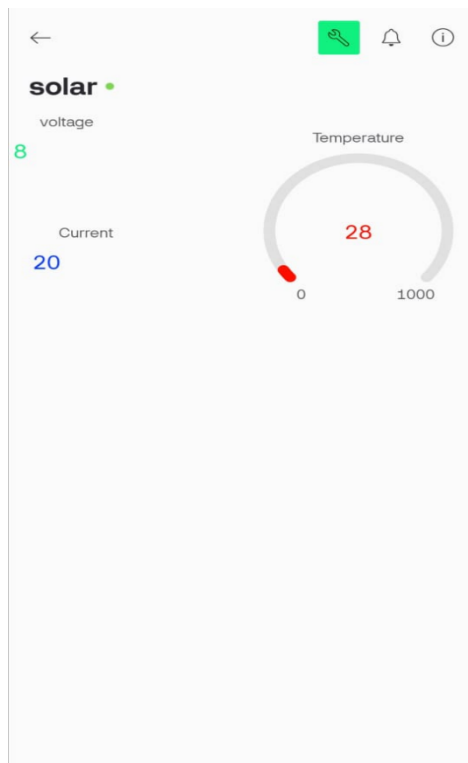
```
}

// Extra loop handler
void extendedLoop()
{
    controlLED();
}

// End of code
```

APPENDIX 2

OUTPUT



REFERENCES

- [1] Yassine Chouay and Mohammed Ouassaid. An intelligent method for fault diagnosis in photovoltaic systems. In 2017 International Conference on Electrical and Information Technologies (ICEIT), pages 1–5. IEEE, 2017.
- [2] Annual Progress Report - AEPC — [aepc.gov.np](https://www.aepc.gov.np/documents/annual-progress-report-aepc). <https://www.aepc.gov.np/documents/annual-progress-report-aepc>. [Accessed 09-09-2023].
- [3] Elias Roumpakias and Tassos Stamatelos. Health monitoring and fault detection in photovoltaic systems in central greece using artificial neural networks. *Applied Sciences*, 12(23):12016, 2022.
- [4] Ahmed A Al-Katheri, Essam A Al-Ammar, Majed A Alotaibi, Wonsuk Ko, Sisam Park, and Hyeong-Jin Choi. Application of artificial intelligence in pv fault detection. *Sustainability*, 14(21):13815, 2022.
- [5] Hee-Won Lim, Il-Kwon Kim, Ji-Hyeon Kim, and UCheul Shin. Simulation-based fault detection remote monitoring system for small-scale photovoltaic systems. *Energies*, 15(24):9422, 2022.
- [6] Siva Ramakrishna Madeti and SN Singh. A comprehensive study on different types of faults and detection techniques for solar photovoltaic system. *Solar Energy*, 158:161–185, 2017.
- [7] C. Packard, J. Wohlgemuth, and S. Kurtz. Development of a Visual Inspection

Data Collection Tool for Evaluation of Fielded PV Module Condition. aug 1 2012.

[8] J. Tsanakas, D. Chrysostomou, P. Botsaris, and A. Gasteratos. Fault diagnosis of photovoltaic modules through image processing and Canny edge detection on field thermographic measurements. *International Journal of Sustainable Energy*, 34(6):351–372, aug 14 2013.

[9] S. R. Madeti and S. Singh. Monitoring system for photovoltaic plants: A review. *Renewable and Sustainable Energy Reviews*, 67:1180–1207, 1 2017.

[10] R. G. Vieira, M. Dhimish, F. M. U. de Araújo, and M. I. S. Guerra. Pv Module Fault Detection Using Combined Artificial Neural Network and Sugeno Fuzzy Logic. *Electronics*, 9(12):2150, dec 15 2020.

[11] M. Hussain, M. Dhimish, S. Titarenko, and P. Mather. Artificial neural network based photovoltaic fault detection algorithm integrating two bi-directional input parameters. *Renewable Energy*, 155:1272–1292, 8 2020.

[12] Paul Stackhouse. NASA POWER | Data Access Viewer — power.larc.nasa.gov. <https://power.larc.nasa.gov/data-access-viewer/>. [Accessed 09-09- 2023].

[13] K. El Bouchefry and R. S. de Souza. Learning in Big Data: Introduction to Machine Learning. *Knowledge Discovery in Big Data from Astronomy and Earth Observation*, pages 225–249, 2020.



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
GOVERNMENT COLLEGE OF TECHNOLOGY, COIMBATORE - 641 013
(An Autonomous Institution Affiliated to Anna University, Chennai - 25)



**7th NATIONAL CONFERENCE ON ADVANCED COMPUTING
AND COMMUNICATION SYSTEMS**

Certificate of Appreciation

This is to certify that Dr./Mr./Ms.. E. Dhivya..... of..Vivakaranha..College..of..Technology..for..women..... has presented a paper titled ..Detection..of..Malfunctioning...Photovoltaic modules..using..ML..and..IoT..... in the 7th National Conference on Advanced Computing and Communication Systems (NCACCS'26) on March 12, 2026 organized by the Department of Electronics and Communication Engineering.


Dr.S.Mathivanan
Asst. Prof./ECE


CO-ORDINATORS
Dr.G.Thirugnanam
Asso. Prof.(CAS)/ECE


Dr.J.Rangaraj
Asst. Prof./ECE


CONVENER
Dr.M.Santhi
HOD/PECE


PATRON
Dr.K.Manomani
Principal



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
GOVERNMENT COLLEGE OF TECHNOLOGY, COIMBATORE - 641 013
(An Autonomous Institution Affiliated to Anna University, Chennai - 25)



**7th NATIONAL CONFERENCE ON ADVANCED COMPUTING
AND COMMUNICATION SYSTEMS**

Certificate of Appreciation

This is to certify that **Dr./Mr./Ms... R. Lavanya**..... has presented a paper titled
of **.Vivekanandha.. College of Technology.. for.. women.....** has presented a paper titled
.. **Detection of Multifunctioning Photovoltaic modules using ML and IoT** ..
in the 7th National Conference on Advanced Computing and Communication Systems
(NCACS'26) on March 12, 2026 organized by the Department of Electronics and
Communication Engineering.

CO-ORDINATORS
Dr.S.Mathivanan
Asst. Prof./ECE

CO-ORDINATORS
Dr.G.Thiruganham
Asso. Prof.(CAS)/ECE

Dr.J.Rangaraj
Asst. Prof./ECE

CONVENER
Dr.M.Santhi
HOD/PECE

PATRON
Dr.K.Manonmani
Principal



(An Autonomous Institution Affiliated to Anna University, Chennai - 25)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
GOVERNMENT COLLEGE OF TECHNOLOGY, COIMBATORE - 641 013

**7th NATIONAL CONFERENCE ON ADVANCED COMPUTING
AND COMMUNICATION SYSTEMS**

Certificate of Appreciation

This is to certify that Dr./Mr./Ms... M. Nehasi.....
of .Vivekanandha.. College. of .Technology. for .woman..... has presented a paper titled
..Detection. of .Malfunctioning.. Photovoltaic. modules. using. ML. and. IoT.....
in the 7th National Conference on Advanced Computing and Communication Systems
(NCACS'26) on March 12, 2026 organized by the Department of Electronics and
Communication Engineering.

Dr.S.Mathivanan
Asst. Prof./ECE

CO-ORDINATORS
Dr.G.Thirugnanam
Asso. Prof.(CAS)/ECE

Dr.J.Rangaraj
Asst. Prof./ECE

CONVENER
Dr.M.Santhi
HOD/PECE

PATRON
Dr.K.Manonmani
Principal



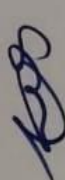
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
GOVERNMENT COLLEGE OF TECHNOLOGY, COIMBATORE - 641 013
(An Autonomous Institution Affiliated to Anna University, Chennai - 25)



**7th NATIONAL CONFERENCE ON ADVANCED COMPUTING
AND COMMUNICATION SYSTEMS**

Certificate of Appreciation

This is to certify that Dr./Mr./Ms. M. Rithika.....
of .Vivekanandha. College. of Technology. for women..... has presented a paper titled
. Datafusion. of. Multifunctioning. Photovoltaic. modules. using. ML. and. IoT.
in the 7th National Conference on Advanced Computing and Communication Systems
(NCACS'26) on March 12, 2026 organized by the Department of Electronics and
Communication Engineering.


CO-ORDINATORS
Dr.S.Mathivanan
Asst. Prof./ECE


Dr.G.Thirugnanam
Asso. Prof.(CAS)/ECE


Dr.J.Rangaraj
Asst. Prof./ECE


CONVENER
Dr.M.Santhi
HOD/PECE


PATRON
Dr.K.Manomani
Principal